

Blood Flow Restriction Training Does Not Improve Quadriceps Strength After Anterior Cruciate Ligament Reconstruction

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Objectives: Blood flow restriction training (BFRT) involves obstruction of venous outflow of working muscles during exercise and may lead to more substantial recovery of quadriceps strength after anterior cruciate ligament (ACL) reconstruction. The purpose of this study is to examine the effects of BFRT in ACL reconstruction patients before and after exposure to BFRT.

Methods: This study was a randomized controlled trial in which 36 patients who had torn their ACL and were scheduled to undergo ACL reconstructive surgery (ACLR) with an autograft were randomized to receive exercise with BFRT (N=18) or exercise without BFRT (N=18). Participants in both groups performed the same exercise, but either did or did not have blood flow restricted. The exercise component of the intervention consisted of all subjects performing a single-leg isokinetic leg press, at an intensity of 70% of the subjects' one-repetition maximum, for four sets of ten repetitions two times per week for 8 weeks beginning at 8 weeks post-operatively. Patients randomized to the BFRT group performed the leg-press exercise with a blood flow restriction cuff applied to the thigh and set to a limb occlusion pressure of 80 percent. All participants were concurrently undergoing standard ACL rehabilitation at the same physical therapy clinic. Bilateral isometric (recorded at a knee angle of 90°) and isokinetic (60°/second) quadriceps strength were recorded using a Biodex dynamometer (System 4, Shirley, NY) prior to ACLR and after the conclusion of the 8-week intervention. Peak isometric and isokinetic quadriceps strength were utilized to generate isometric and isokinetic quadriceps symmetry indices ((ACLR limb strength/Healthy limb strength) x100). Change from baseline symmetry scores were then generated using the following equation: (post-intervention symmetry - pre-intervention symmetry)/pre-intervention symmetry in order to account for possible differences in strength between groups prior to intervention delivery. Change from baseline isometric and isokinetic quadriceps symmetry scores were then compared between groups (exercise with BFRT, exercise without BFRT) using one-way analysis of variance tests with an a priori α set to $P \leq 0.05$. Effect sizes (Cohen's d) and 95% confidence intervals were also computed.

Results: No significant differences were found for change from baseline isokinetic quadriceps symmetry index ($P=0.39$, BFRT mean=-0.05, Control mean=-0.19) or change from baseline isometric quadriceps symmetry index ($P=0.62$, BFRT mean=-0.04, control mean=-0.10). The effect sizes for isokinetic quadriceps symmetry index ($d=0.28$, 95% CI= -0.37, 0.93) and isometric quadriceps symmetry ($d=0.16$, 95% CI= -0.49, 0.80) were small with confidence intervals that crossed zero.

Conclusion: An 8-week blood flow resistance training + exercise intervention did not increase quadriceps muscle strength in patients who had undergone ACL reconstruction. Based on our findings, application of blood flow restriction training in ACL reconstruction patients to improve quadriceps strength may not be warranted. Future studies may benefit from a longer follow-up and larger sample size.

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